

## EDUCATION

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### University of California Santa Barbara

Ph.D. in Geography, Advisor: Leila Carvalho

Santa Barbara, CA

2017–June 2022

- Dissertation: “Dynamics of Atmospheric Rivers in High Mountain Asia: Influences on Precipitation, Lightning and Landslides”

### California State University, Los Angeles

M.A. in Geography, GPA: 4.0

Los Angeles, CA

2015–2017

- Thesis: “Atmospheric River Contributions to Extratropical Poleward Moisture Transports and Atmospheric Water Cycle”

### University of Colorado Colorado Springs

B.A. in Geography, GPA: 3.8

Colorado Springs, CO

2010–2014

- Emphasis in Geospatial Science with Certification in Geographic Information Systems
- Thesis: “Examining post-fire landscape change using remote sensing”
- UCCS Department of Geography Honors with highest distinction

### University of Granada

Study Abroad

Granada, Spain

2011–2012

### University of Costa Rica

Study Abroad

San Jose, Costa Rica

2014–2014

## PUBLICATIONS

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- McGuire, Luke, Nina Oakley, and **Deanna Nash** (2026). “Mesoscale Model Performance for Short-Duration Rainfall During Postfire Flood and Debris-Flow Events in California”. In: *in preparation*.
- Nash, Deanna**, Jon Rutz, and Aaron Jacobs (2026). “Near-Future Increases in Extreme Precipitation and Rain-on-Snow Events in Southeast Alaska”. In: *in review Journal of Geophysical Research: Atmospheres*. DOI: <https://doi.org/10.22541/essoar.176797933.33867091/v1>.
- Nash, Deanna**, Jon Rutz, Aaron Jacobs, and Brian Kawzenuk (2026). “Using Model Climate to improve situational awareness ahead of atmospheric river-related landslide events in Southeast Alaska”. In: *in press Journal of Operational Meteorology*.
- Battula, Suma, Jason Cordeira, **Deanna Nash**, Jon Rutz, and F. Martin Ralph (2025). “Leveraging the Atmospheric River Framework to Categorize Top-Decile Precipitation Regimes in Colorado”. In: *Geophysical Research Letters*. DOI: <https://doi.org/10.1029/2025GL117528>.
- Nash, Deanna**, Jon Rutz, Jason Cordeira, Zhenhai Zhang, F. Martin Ralph, Kris Sanders, and Erin Walter (2025). “A Trajectory-based Method for Estimating the Contribution of Landfalling Atmospheric Rivers to Top-Decile Precipitation Across Colorado”. In: *Journal of Geophysical Research: Atmospheres*. DOI: <https://doi.org/10.1029/2025JD043580>.
- Oakley, Nina, Derek Cheung, Don Lindsay, and **Deanna Nash** (2025). “Insights from a 25-year database of postfire debris flows in California”. In: *International Journal of Wildland Fire*. DOI: <https://doi.org/10.1071/WF25136>.
- Nash, Deanna**, Jon Rutz, and Aaron Jacobs (2024). “Atmospheric Rivers In Southeast Alaska: Meteorological Conditions Associated With Extreme Precipitation”. In: *Journal of Geophysical Research: Atmospheres*. DOI: <https://doi.org/10.1029/2023JD039294>.

- Wang, Zhibiao, Qinghua Ding Ding, Renguang Wu, Thomas J. Ballinger, Bin Guan, Deniz Bozkurt, **Deanna Nash**, Ian Baxter, Dániel Topál, Zhe Li, Gang Huang, Wen Chen, Shangfeng Chen, Xi Cao, and Zhang Chen (2024). “Role of atmospheric rivers in shaping long term Arctic moisture variability”. In: *Nature Communications*. DOI: <https://doi.org/10.1038/s41467-024-49857-y>.
- Nash, Deanna**, Leila M. V. Carvalho, Jon Rutz, and Charles Jones (2023). “Influence of the freezing level on Atmospheric Rivers in High Mountain Asia: WRF case studies of orographic precipitation extremes”. In: *Climate Dynamics*. DOI: <http://dx.doi.org/10.1007/s00382-023-06929-x>.
- Murray, Alan T, Leila Carvalho, Richard L Church, Charles Jones, Dar Roberts, Jing Xu, Katelyn Zigner, and **Deanna Nash** (2021). “Coastal Vulnerability under Extreme Weather”. In: *Applied Spatial Analysis and Policy* 14.3, pp. 497–523. ISSN: 1874-4621. DOI: 10.1007/s12061-020-09357-0.
- Nash, Deanna**, Leila M. V. Carvalho, Charles Jones, and Qinghua Ding (2021). “Winter and spring atmospheric rivers in High Mountain Asia: climatology, dynamics, and variability”. In: *Climate Dynamics*. ISSN: 1432-0894. DOI: 10.1007/S00382-021-06008-Z.
- Nash, Deanna** and Leila Carvalho (2020). “Brief Communication: An electrifying atmospheric river—understanding the thunderstorm event in Santa Barbara County during March 2019”. In: *Natural Hazards and Earth System Sciences* 20.7, pp. 1931–1940. DOI: <https://doi.org/10.5194/nhess-20-1931-2020>.
- Nash, Deanna**, Duane Waliser, Bin Guan, Hengchun Ye, and F Martin Ralph (2018). “The role of atmospheric rivers in extratropical and polar hydroclimate”. In: *Journal of Geophysical Research: Atmospheres* 123.13, pp. 6804–6821. DOI: <https://doi.org/10.1029/2017JD028130>.
- Nash, Deanna**, Hengchun Ye, and Eric Fetzer (2017). “Spatial and Temporal Variability in Winter Precipitation across the Western United States during the Satellite Era”. In: *Remote Sensing* 9.9, p. 928. DOI: <https://doi.org/10.3390/rs9090928>.

## SELECTED PRESENTATIONS

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- Nash, Deanna** (Mar. 2026a). “Atmospheric Rivers, Flooding, and the Changing Climate in the San Diego Region (invited oral)”. In: *City of San Diego Resilience Board Meeting*. San Diego, CA.
- Nash, Deanna** (June 2026b). “The Center for Western Weather and Water Extremes and Forecast Informed Reservoir Operations (invited oral)”. In: *National Hydropower Association Meeting*. La Jolla, CA.
- Nash, Deanna**, A. Jacobs, J. Rutz, and B. Kawzenuk (Mar. 2026). “Atmospheric River Controls on Extreme Rainfall and Landslide Hazard in Southeast Alaska (invited oral)”. In: *LandAware Webinar Series*. Virtual.
- Nash, Deanna** (July 2024a). “Atmospheric Rivers and the Climate Crisis: Changes in Extreme Precipitation in the Western U.S. (invited oral)”. In: *Global Leadership Institute Future Leaders Summer Program*. La Jolla, CA.
- Nash, Deanna** (Oct. 2024b). “Atmospheric Rivers in San Diego: Historical Flooding in January 2024 and Increasing Drought Resilience (invited oral)”. In: *UC San Diego Community Advisory Board Meeting*. La Jolla, CA.
- Nash, Deanna**, A. Jacobs, and J. Rutz (Aug. 2024). “Differentiating between impactful and non-impactful Atmospheric River events in Southeast Alaska (oral)”. In: *Virtual Alaska Weather Symposia*. Virtual.
- Nash, Deanna**, J. Rutz, J. Cordeira, F.M. Ralph, K. Sanders, and E. Walter (May 2024). “Historical Precipitation Contribution of Inland-Penetrating Atmospheric Rivers in the Upper Yampa (invited oral)”. In: *Yampa Basin Rendezvous*. Steamboat Springs, CO.
- Nash, Deanna** (July 2023a). “Atmospheric Rivers and the Climate Crisis: Changes in Extreme Precipitation in the Western U.S. (invited oral)”. In: *Global Leadership Institute Future Leaders Summer Program*. La Jolla, CA.
- Nash, Deanna** (May 2023b). “Extremes in the Atmosphere, Disasters on Land: Using Dynamically Downscaled Climate Reanalysis To Understand Atmospheric River-Induced Extreme Precipitation In Southeast Alaska (invited oral)”. In: *Center for Land-Surface Hazards Modeling Expo*. Virtual.

- Nash, Deanna**, J. Cordeira, B. Kawzenuk, S. Bartlett, L. Bosart, C. Castellano, A. Cobb, C. Hecht, T.Y. Hsu, J. Kalansky, T. Leicht, A. Mitchell, B. Moore, and Zou X. (Oct. 2023). "Climatological Perspective and Synoptic Analysis of Landfalling Atmospheric Rivers in California during December 2022 and January 2023 (invited oral)". In: *Geological Society of America Connects*. Pittsburgh, PA.
- Nash, Deanna**, Jon Rutz, and Aaron Jacobs (Sept. 2023). "Atmospheric Rivers in Southeast Alaska: Meteorological Conditions Associated With Extreme Precipitation (invited oral)". In: *University of Alaska Southeast Natural History Seminar*. Sitka, AK.

## EXPERIENCE

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| <b>Center for Western Weather and Water Extremes</b><br><b>Scripps Institution of Oceanography, University of California San Diego</b><br>Precipitation and Geohazard Scientist                                                                                                                                                    | San Diego, CA<br>August 2025–present                 |
| <ul style="list-style-type: none"><li>– Extreme precipitation from atmospheric rivers that trigger geohazards</li></ul>                                                                                                                                                                                                            |                                                      |
| <b>Center for Western Weather and Water Extremes</b><br><b>Scripps Institution of Oceanography, University of California San Diego</b><br>Postdoctoral Scholar                                                                                                                                                                     | San Diego, CA<br>August 2022–August 2025             |
| <ul style="list-style-type: none"><li>– Atmospheric Rivers in Southeast Alaska</li><li>– National Science Foundation Award: 2052972</li><li>– <u>Kutí</u> Project: Understanding Natural Hazards and Supporting Community Response</li></ul>                                                                                       |                                                      |
| <b>University of California Santa Barbara</b><br>Field Researcher                                                                                                                                                                                                                                                                  | Santa Barbara, CA<br>April 2022 –May 2022            |
| <ul style="list-style-type: none"><li>– Sundowner Winds Experiment (SWEX)</li><li>– improved understanding of the dynamics and predictability of downslope windstorms in coastal Santa Barbara County</li><li>– launched 3-hourly radiosondes during significant Sundowner events</li></ul>                                        |                                                      |
| <b>National Center for Atmospheric Research</b><br>Advanced Student Program Colloquium                                                                                                                                                                                                                                             | Boulder, CO<br>July 2021                             |
| <ul style="list-style-type: none"><li>– The Science of Seasonal to Subseasonal Predictions</li><li>– Group Leads: Aneesh Subramanian and Mike DeFlorio</li><li>– Used python to perform S2S hindcast evaluation on Atmospheric Rivers in Western US</li></ul>                                                                      |                                                      |
| <b>Jet Propulsion Laboratory</b><br>Intern Earth Sciences Division                                                                                                                                                                                                                                                                 | Pasadena, CA<br>October 2015 –September 2017         |
| <ul style="list-style-type: none"><li>– Regional Climate Model Evaluation Systems</li><li>– Mentor: Duane Waliser</li><li>– Used python to help implement software for comparing regional climate models</li><li>– Maintained the website with HTML and CSS (<a href="http://rcmes.jpl.nasa.gov">rcmes.jpl.nasa.gov</a>)</li></ul> |                                                      |
| <b>California State University Los Angeles</b><br>Graduate Assistant Geosciences Lab                                                                                                                                                                                                                                               | Los Angeles, CA<br>October 2015 –June 2016           |
| <ul style="list-style-type: none"><li>– Assisted students with GIS needs</li></ul>                                                                                                                                                                                                                                                 |                                                      |
| <b>University of Colorado</b><br>Research Assistant Department of Geography                                                                                                                                                                                                                                                        | Colorado Springs, CO<br>September 2014 –January 2015 |
| <ul style="list-style-type: none"><li>– Post-fire vegetation regrowth</li></ul>                                                                                                                                                                                                                                                    |                                                      |

- Performed geospatial analysis with remote sensing imagery
- Experience in python and IDRISI

### Colorado Springs Fire Department

GIS Technician Division of the Fire Marshal

Colorado Springs, CO

July 2013 –January 2014

- Created maps and surveyed wildfire mitigation project areas
- Collected weekly fuels samples to measure fire risk
- Created and maintained web map for Wildland Urban Interface

### City of Colorado Springs

GIS Technician Information Technology

Colorado Springs, CO

January 2014 –June 2014

- Developed maps for different city divisions
- Worked in a versioned geodatabase environment
- Experience in python, geocoding, topology editing, linear referencing, and other geoprocessing tools

## TEACHING

|                                                                                                                                                  |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| • <b>Instructor of Record</b> at University of California Santa Barbara<br><i>Introduction to Meteorology (GEOG110)</i>                          | Summer 2020 |
| • <b>Session Lead</b> at University of California Santa Barbara Library Collaboratory<br><i>Ecodatascience session on Python dask and xarray</i> | Spring 2019 |
| • <b>Instructor of Record</b> at University of California Santa Barbara<br><i>Waves and Tides in the Ocean (GEOG165)</i>                         | Summer 2018 |
| • <b>Teaching Associate</b> at California State University Los Angeles<br><i>Introduction to Physical Geography (GEOG1600)</i>                   | Spring 2017 |
| • <b>Teaching Associate</b> at California State University Los Angeles<br><i>Introduction to Physical Geography (GEOG1600)</i>                   | Fall 2016   |

## SKILLS

- **High Performance Computing:** cluster management tools like Slurm and PBS
- **Git and Github:** create and manage versions of code repositories
- **Modeling Software:** Advanced Research Weather Research and Forecasting Model
- **Geospatial Software:** ArcGIS and QGIS
- **Remote Sensing Software:** IDRISI and ENVI
- **Language:** Spanish (intermediate)

## LANGUAGES

- **Python:** advanced
- **HTML/CSS/JavaScript:** advanced
- **R:** advanced
- **Matlab:** advanced
- **bash:** advanced
- **IDL:** intermediate

## PEER REVIEW

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|----------------------------------------------------------------|--------------------------------------------------------|
| • <i>Geophysical Research Letters</i>                          | June 2026                                              |
| • <i>Journal of Geophysical Research - Atmospheres</i>         | April 2021, March 2023, Aug 2023, April 2024, Feb 2025 |
| • <i>Journal of Climate</i>                                    | Jan 2024                                               |
| • <i>Earth System Science Data</i>                             | March 2021                                             |
| • <i>Environmental Research Communications</i>                 | June 2020                                              |
| • <i>Quarterly Journal of the Royal Meteorological Society</i> | Oct 2019                                               |
| • <i>Portuguese Polar Program (PROPOLAR) Project Proposals</i> | Aug 2019, April 2021                                   |

## SCHOLARSHIPS AND AWARDS

|                                                                          |            |
|--------------------------------------------------------------------------|------------|
| • Sitka Sound Science Center Scientist in Residency Fellowship           | 2023, 2025 |
| • UCSB Department of Geography Excellence in Research Award              | 2022       |
| • New Frontiers Graduate Fellow - NSF awards OCI-0725070 and ACI-1238993 | 2021–22    |
| • NASA Earth and Space Science Fellowship #80NSSC18K1412                 | 2018–21    |
| • AAG Climate Specialty Group Student Paper Competition: 2nd place       | April 2019 |
| • Regents Fellowship UCSB                                                | 2017–18    |
| • NASA DIRECT-STEM MIRO #NNX15AQ06A                                      | 2015–17    |
| • CSULA Gamma Theta Upsilon Scholarship                                  | 2016–17    |
| • John David Rees Research Scholarship                                   | 2016–17    |
| • UCCS Letters, Arts, and Sciences Research Grant                        | 2013–14    |
| • UCCS Women in Geography Award                                          | 2013–14    |
| • UCCS Honors Scholarship Program                                        | 2010–14    |
| • UCCS Reach Your Peak Scholarship Program                               | 2010–14    |
| • UCCS Deans and Presidents List                                         | 2010–14    |

## SERVICE

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| • Co-Chair for the Yampa Basin Rendezvous<br><i>Center for Western Weather and Water Extremes, Scripps Institute of Oceanography, UCSD</i>                          | 2026                     |
| • Steering Committee Member for the Yampa Basin Rendezvous<br><i>Center for Western Weather and Water Extremes, Scripps Institute of Oceanography, UCSD</i>         | 2025                     |
| • Member of the CW3E Equity, Diversity, and Inclusion Task Force<br><i>Center for Western Weather and Water Extremes, Scripps Institute of Oceanography, UCSD</i>   | Winter 2023–Winter 2026  |
| • Co-chair of the CW3E Equity, Diversity, and Inclusion Task Force<br><i>Center for Western Weather and Water Extremes, Scripps Institute of Oceanography, UCSD</i> | Winter 2024–Winter 2025  |
| • Advanced Graduate Student Mentor<br><i>UCSB Graduate Scholars Program</i>                                                                                         | Fall 2021–Spring 2022    |
| • oSTEM Mentor<br><i>out in STEM UCSB chapter</i>                                                                                                                   | Winter 2021–Spring 2021  |
| • Alumnae Advisory Committee Recruitment Advisor<br><i>CA Zeta chapter of Pi Beta Phi</i>                                                                           | Fall 2019–Fall 2022      |
| • Chair of the Graduate Advisory Committee<br><i>Geography Department, UCSB</i>                                                                                     | Fall 2019–June 2022      |
| • Chair of the Lounge Committee<br><i>Geography Department, UCSB</i>                                                                                                | Fall 2019–Spring 2020    |
| • Member of the Events Committee<br><i>Department of Geography, UCSB</i>                                                                                            | Fall 2017 –Fall 2019     |
| • Geography Awareness Week Elementary School Outreach<br><i>Geography Department, UCSB</i>                                                                          | Fall 2018, '19, '20      |
| • President of the Lambda Pi Chapter of Gamma Theta Epsilon Honor Society<br><i>Department of Geography, CSULA</i>                                                  | Spring 2016 –Spring 2017 |
| • Member of the Theta Chi Chapter of Gamma Theta Epsilon Honor Society<br><i>Department of Geography, UCCS</i>                                                      | Spring 2014              |

- Member of the Colorado Epsilon Chapter of Pi Beta Phi  
*University of Colorado Colorado Springs*

Fall 2011 –Spring 2014